



Non-Genetically Modified Adoptive Cell Therapies for Solid Tumors: Current Landscape and Future Challenges

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Received: 28 February 2025 / Accepted: 9 December 2025
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Abstract

Non-genetically modified adoptive cell therapies (ACTs) represent a rapidly advancing frontier in solid tumor immunotherapy, offering a safe and adaptable alternative to genetically engineered approaches by capitalizing on the intrinsic plasticity of immune cells. Genetic engineering strategies, including CAR-T cells, encounter significant obstacles in solid tumors, including on-target off-tumor toxicity, an immunosuppressive tumor microenvironment, and drug resistance. Although non-genetically modified ACTs—including tumor-infiltrating lymphocytes (TILs), cytokine-induced killer (CIK) cells, natural killer (NK) cells, and $\gamma\delta$ T cells—offer unique advantages, their clinical application remains underexplored. This review consolidates the mechanistic basis, clinical progress, and limitations of non-genetically modified ACTs, proposing a paradigm shift toward combinatorial strategies. We systematically assessed how TILs overcome tumor microenvironment (TME) inhibition through lymphodepletion and cytokine assistance, compared the histocompatibility complex-unrestricted cytotoxicity of CIK cells with their functional diversity, and emphasized the innate flexibility of NK/ $\gamma\delta$ T cells against antigen-loss variants. By integrating preclinical and clinical data, we identify critical challenges: in vitro expansion inefficiency, absence of standardized protocols, and dynamic TME interactions. Furthermore, we advocated patient stratification by biomarkers, the addition of optimized cytokines, and rational combinations with checkpoint inhibitors or metabolic modulators to enhance efficacy. This review outlines the current landscape and proposes actionable solutions to reconcile the disparity between experimental potential and clinical applicability in non-genetically modified ACT.

Keywords Adoptive cell therapy · Solid tumor treatment · Non-genetically modified · Tumor-infiltrating lymphocytes · Tumor microenvironment

Abbreviations

ACT Adoptive cell therapy
 ADC Antibody–drug conjugate

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ADCC	Antibody-dependent cell-mediated cytotoxicity
AE	Adverse event
AG1	G6PD activator
APC	Antigen-presenting cells
CAR	Chimeric antigen receptor
CIK	Cytokine-induced killer cells
CR	Complete response
CRS	Cytokine release syndrome
CT	Computed Tomography
CTL	Cytotoxic T lymphocyte
CTLA-4	Cytotoxic T lymphocyte associate protein-4
DC	Dendritic cells
DFS	Disease-free survival
DNAM-1	DNAX accessory molecule 1
EGFR	Epidermal growth factor receptor
FAS/FASL	Factor-related apoptosis/factor-related apoptosis ligand
FDA	US Food and Drug Administration
GvHD	Graft versus host disease
HCC	Hepatocellular carcinoma
HER2	Human epidermal growth factor receptor-2
HLA	Human leukocyte antigen
ICIs	Immune-checkpoint inhibitors
IFN- γ	Interferon- γ
IL-2	Interleukin-2
iNKT	Invariant natural killer T cells
IPP	Isopentenyl pyrophosphate
LAK	Lymphokine-activated killer cell
mCRC	Metastatic Colorectal cancer
MHC	Major histocompatibility complex
mNSCLC	Non-small-cell lung cancer
NK	Natural killer cells
NKG2D	Natural killer group 2 member D
NKR	Natural Killer Cell Receptor
NMA-LD	Non-myeloablative lymphodepleting
OKT3	Anti-CD3 monoclonal antibody
ORR	Objective response rate
OS	Overall survival
OTOT	On target, off-tumor toxicity
PBLs	Peripheral blood lymphocytes
PBMC	Peripheral blood mononuclear cell
PD-1	Programmed cell death-1
PD-L1	Programmed cell death-Ligand 1
PFS	Progression-free survival
PPP	Pentose phosphate pathway
RCT	Randomized-controlled trial
rhIL-2	Recombinant human interleukin-2
SCID	Server combined immune-deficiency
TACE	Transarterial chemoembolization
TAE	Transcatheter arterial embolization
TBI	Total body irradiation
TCR	T-cell receptor

TEAE	Treatment Emergent Adverse Event
Th1	Helper T cell
TILs	Tumor-infiltrating lymphocytes
TNBC	Triple negative breast cancer
TNF- α	Tumor necrosis factor alpha;
TRAIL	Tumor-necrosis factor related apoptosis-inducing ligand
Treg	Regulatory T cells
XELOX	Capecitabine plus oxaliplatin

Introduction

Solid tumors remain difficult to treat with immunotherapy because of their immunosuppressive tumor micro-environment (TME), extensive antigenic heterogeneity, and structural barriers that impede effective immune-cell infiltration. Conventional treatment options, including chemotherapy and radiotherapy, although somewhat effective in managing tumor progression, frequently entail high rates of medical complications, adverse effects, and diminished life expectancy[1]. Although genetically engineered adoptive cell therapies (ACTs; chimeric antigen receptor T-cell [CAR-T]) have revolutionized hematological malignancies, their efficacy in solid tumors remains limited by toxicity, antigenic escape, and T-cell exhaustion. Genetically engineered ACT includes T-cell receptor (TCR)-T-cell therapy, which employs naturally occurring TCRs that target intracellular antigens in a histocompatibility complex (MHC)-restricted fashion. Although CAR-T therapy encounters considerable challenges in solid tumors, TCR-T therapy has demonstrated potential in targeting certain shared tumor antigens; however, it is limited by human leukocyte antigen restriction and the availability of appropriate TCRs [2, 3]. Moreover, although immune checkpoint inhibitors (ICIs) have demonstrated notable success in some solid tumors, their efficacy can be limited in ‘immune-cold’ tumors, typically defined by an absence of pre-existing T-cell infiltration[4]. Non-genetically modified ACTs, utilizing naturally occurring or ex vivo-activated immune cells, present a compelling alternative [5]. These therapies circumvent the risks associated with permanent genetic modification, including genotoxicity, insertional oncogenesis, and the potential of generating autoreactive T cells [6]. This review examines a significant knowledge deficiency: how may non-genetically modified ACTs be enhanced to surmount the distinct challenges posed by solid tumors? We evaluated three dimensions: mechanism-driven efficacy, clinical scalability, and synergy with multimodal therapies. We proposed a combination of ACTs and investigate the role of cellular metabolism and TME reprogramming on ACTs.

Brief History of Non-Genetically Modified Adoptive Cell Therapies

Over the past half-century, the advancement and refinement of immunological cell-based research have propelled tumor immunology into a phase of comprehensive development. In 1968, Miller recognized the essential function of T cells in immunity [7]. Dendritic cells (DCs) were identified in 1973[8]. The identification of natural killer (NK) cells, lymphokine-activated killer (LAK) cells, $\gamma\delta$ T, and natural killer T (NKT) cells has progressively enhanced the cellular basis of tumor immunology [9–12]. A transformative shift from basic science to therapeutic application began in the 1980s with the pioneering work of Rosenberg and colleagues. After establishing that LAK cells could mediate tumor regression, they made the crucial observation that tumor-infiltrating lymphocytes (TILs) expanded in interleukin-2 (IL-2) exhibited dramatically enhanced antitumor activity compared to LAK cells [13]. This discovery rapidly translated into clinical practice, with Topalian et al. demonstrating in 1988 that TIL therapy could mediate tumor regression in patients with metastatic melanoma [14]. The 1990s marked the emergence of additional cellular platforms. Cytokine-induced killer

(CIK) cells were first described by Schmidt-Wolf et al. in 1991[15] and subsequently shown to possess superior antitumor activity compared to LAK cells [16]. The clinical translation of CIK cells was initiated by the end of the decade, establishing their safety and feasibility [17]. Parallel developments included the first clinical application of dendritic cell (DC) vaccines by Hsu et al.[18] and the demonstration of antitumor effects by human $\gamma\delta$ T cells in mouse models[19]. As tumor immunology advances, studies have demonstrated that co-culturing DCs with CIK cells can enhance the cytotoxic activity of CIK cells [20]. Numerous teams have examined and confirmed the safety and viability of reinfusing allogeneic NK cells or invariant natural killer T (iNKT) cells, which also demonstrate antitumor effects [21, 22]. In 2007, Hirohito et al. commenced the reinfusion of $\gamma\delta$ T cells for treating patients with advanced renal cell carcinoma, exhibiting good tolerance and antitumor efficacy [23]. The Food and Drug Administration (FDA) has approved several cell treatments because of advances in immunotherapy [24–28]. In 2024, the FDA approved the first TIL cell therapy, lifileucel, for melanoma treatment [29](Fig. 1). The subsequent section elaborates on the advancement of non-genetically modified adoptive immune-cell therapies.

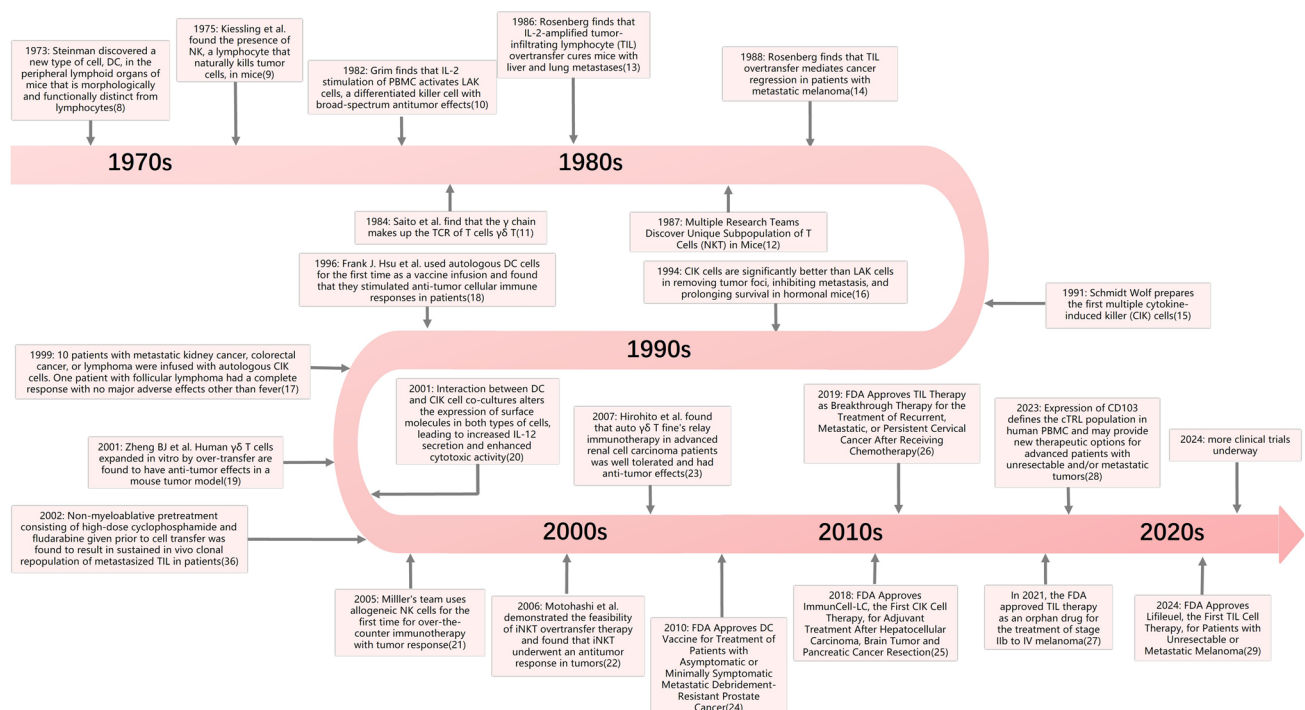


Fig. 1 Timeline for the development of overdose immune-cell therapies. Summary of key landmark studies and breakthroughs in various types of non-gene-edited overdose immune-cell therapies. TIL, tumor-infiltrating lymphocytes; LAK, lymphokine-activated killer

cell; CIK, cytokine-induced killer cells; NK, natural killer cells; iNKT, invariant natural killer T cells; DC, dendritic cells; NMA-LD, non-myeloablative lymphodepletion; FDA, US Food and Drug Administration

Progress in Non-Genetically Modified Adoptive Cell Therapies

Tumor-Infiltrating Lymphocytes

TILs are lymphocytes infiltrating the TME. They migrate from the blood to the TME [30]. They include cytotoxic T cells, helper T cells, regulatory T cells, $\gamma\delta$ T cells, NK cells, B cells, and innate lymphoid cells, all of which contribute to the tumor immune response directly or indirectly [31]. In animal models, CTLs are the primary effector cells that mediate tumor rejection through adoptive transfer of TIL treatment [32]. CTLs induce target cell death by releasing inflammatory cytokines, such as tumor-necrosis factor- α (TNF- α) and Interferon- γ (IFN- γ); apoptosis-inducing proteins, including FAS ligand (FASL) and TNF-related apoptosis-inducing ligand (TRAIL); and cytotoxic granules, leading to perforin-mediated lysis (Fig. 2) [33]. Carsten et al. discovered that an increase in stromal TILs correlated with longer progression-free survival (PFS) and overall survival (OS) in patients with triple-negative breast cancer (TNBC) and HER2-positive breast cancer receiving neoadjuvant chemotherapy [34]. The quantity and functional state of TILs correlate with prognosis in patients with tumors, especially in melanoma, non-small cell lung cancer, breast cancer, ovarian cancer, and head and neck cancer [35–37]. Park et al. posited that stromal TILs are important and independent prognostic markers for untreated patients with early TNBC [36]. These findings offer a theoretical foundation for TIL therapy.

TILs can be extracted from excised tumor tissues and cultured in vitro. Tumor tissue, typically obtained from subcutaneous sites or lymph nodes, is digested or fragmented and cultured in a medium containing IL-2. TILs were extracted and concentrated after adding anti-CD3 monoclonal antibodies, allogeneic feeder cells, and IL-2 to promote T-cell proliferation. Before reinfusion of the TIL product, the patients underwent nonmyeloablative lymphocyte depletion to enhance the infusion environment (Fig. 3). In 1986, Rosenberg proposed TIL therapy, based on the tumor-eradicating properties of LAK therapy. Research indicates that TILs expanded in IL-2, when reintroduced into tumor-bearing mice, exhibited superior therapeutic efficacy than that of LAK cells [38, 39]. Moreover, the combination therapy involving cyclophosphamide, IL-2, and adoptive TIL improved the therapeutic efficacy [13]. In 1988, Topalian et al. conducted a pilot clinical study utilizing TIL therapy combined with recombinant IL-2 for immunotherapy of patients with advanced cancer. Of the six patients with melanoma, five exhibited TIL-mediated specific lytic activity against their autologous tumor targets. Tumor regression was noted at different sites in various patients, and the toxicity profile resembled that of IL-2, thereby confirming the safety and feasibility of TIL therapy [14]. Rosenberg

et al. reported that nonmyeloablative lymphodepleting (NMA-LD) chemotherapy, comprising cyclophosphamide and fludarabine before TIL therapy, may enhance treatment outcomes [40]. Dudley et al. conducted a study that included 13 patients with metastatic melanoma, who underwent 7 days of lymphodepletion chemotherapy with cyclophosphamide and fludarabine before receiving adoptive T-cell immunotherapy. Functional antitumor lymphocytes undergo rapid proliferation, exhibit enhanced persistence, and mediate substantial tumor destruction [41]. NMA-LD increases serum levels of cytokines IL-7 and IL-5, depletes Treg cell-mediated immune suppression, and eliminates competing immune subsets that consume these cytokines, thus boosting CD8⁺ T-cell effector function [42, 43].

The conventional NMA-LD protocol for TIL therapy presently comprises 60–120 mg/kg cyclophosphamide and 75–125 mg/m² fludarabine [44]. The original TIL therapy necessitates isolating T cells from excised tumor tissue, their expansion in IL-2, and subsequent expansion following tumor reactivity selection. Prolonged in vitro culturing can lead to TIL exhaustion. Rosenberg et al. developed a "young TIL" pre-REP approach that eliminates the dependency on IFN- γ -based tumor reactivity selection, significantly reducing laboratory time, rendering it applicable for a wider patient population, with similar clinical responses [45, 46]. The concurrent treatment of various cytokines can enhance TIL proliferation in some solid tumors, thereby increasing CD8⁺ T-cell percentage.

Lifileucel, an autologous TIL treatment for melanoma developed by Iovance Biotherapeutics, Inc. Based on the positive results of the C-144-01 trial, 153 patients with advanced melanoma who progressed after receiving ICIs and targeted therapies were treated with lifileucel, indicating an objective response rate (ORR) of 31.4%, with 41.7% of responses lasting ≥ 18 months [47]. The FDA has approved lifileucel for the treatment of patients with advanced melanoma who have progressed after ICI and targeted therapy [29]. In another phase II multicenter study, lifileucel was utilized to treat 28 patients with refractory metastatic non-small cell lung cancer (mNSCLC) who had failed previous treatments, yielding an overall response rate (ORR) of 21.4%, including responses in patients who were resistant to standard immunotherapy. Treatment-emergent adverse events (TEAEs) are associated with NMA-LD, IL-2 infusion, and late-stage disease progression, all of which are expected and manageable [48].

TIL therapy has exhibited preliminary efficacy in other solid tumors, including breast, cervical, pancreatic, and colorectal cancers [49, 50]. TIL therapy has demonstrated substantial efficacy and manageable safety in patients with advanced solid tumors, especially in those with cancers resistant to conventional treatments. In TIL therapy, the clinical antitumor response is positively correlated with the total

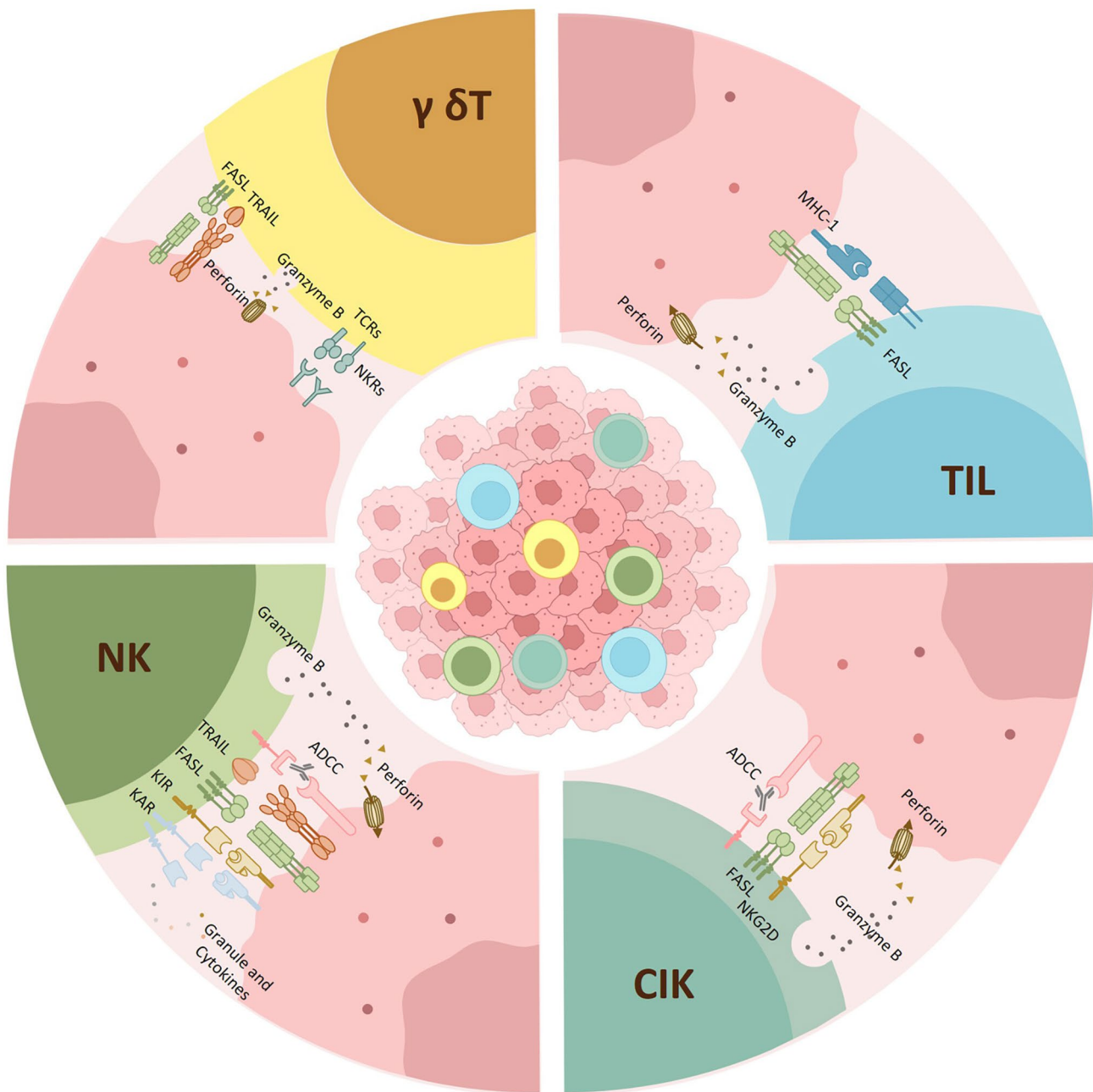


Fig. 2 Killing mechanisms of tumor cells by diverse immune cells. TILs induce target cells by releasing inflammatory cytokines (TNF- α and IFN- γ), apoptosis-inducing proteins (FASL and TRAIL), and cytotoxic degranulation, leading to perforation-mediated lysis. CIK cells can kill tumor cells directly through granzyme- and perforin-mediated cytotoxicity, indirectly by secreting cytokines, such as IFN- γ , TNF- α , and IL-2 to regulate the body's immune response, and inducing apoptosis of tumor cells by binding to FasL on the surface of tumor cells. NK cells release perforin and granzyme to kill tumor cells directly, kill tumor cells via ADCC, produce cytokines and chemokines to modulate adaptive immune responses, and enhance the activity of other immune cells. $\gamma\delta T$ cells kill tumor cells through

ADCC and perforin-granzyme, as well as apoptosis mediated by FasL and TRAILR, and exert a killing effect on a variety of solid tumors through the expression of NKRs and specific recognition of IPPs. $\gamma\delta T$ also produces cytokines, such as IFN- γ , TNF- α , and IL-2, which promote the proliferation and differentiation of CD8⁺T cells and enhance cellular antitumor activity. TILs, tumor-infiltrating lymphocytes; CIK, cytokine-induced killer cells; NK, natural killer cells; IFN- γ , interferon- γ ; TNF- α , tumor-necrosis factor alpha; FASL, factor-related apoptosis-inducing ligand; TRAIL, tumor-necrosis factor related apoptosis-inducing ligand; ADCC, antibody-dependent cell-mediated cytotoxicity; IPP, isopentenyl pyrophosphate

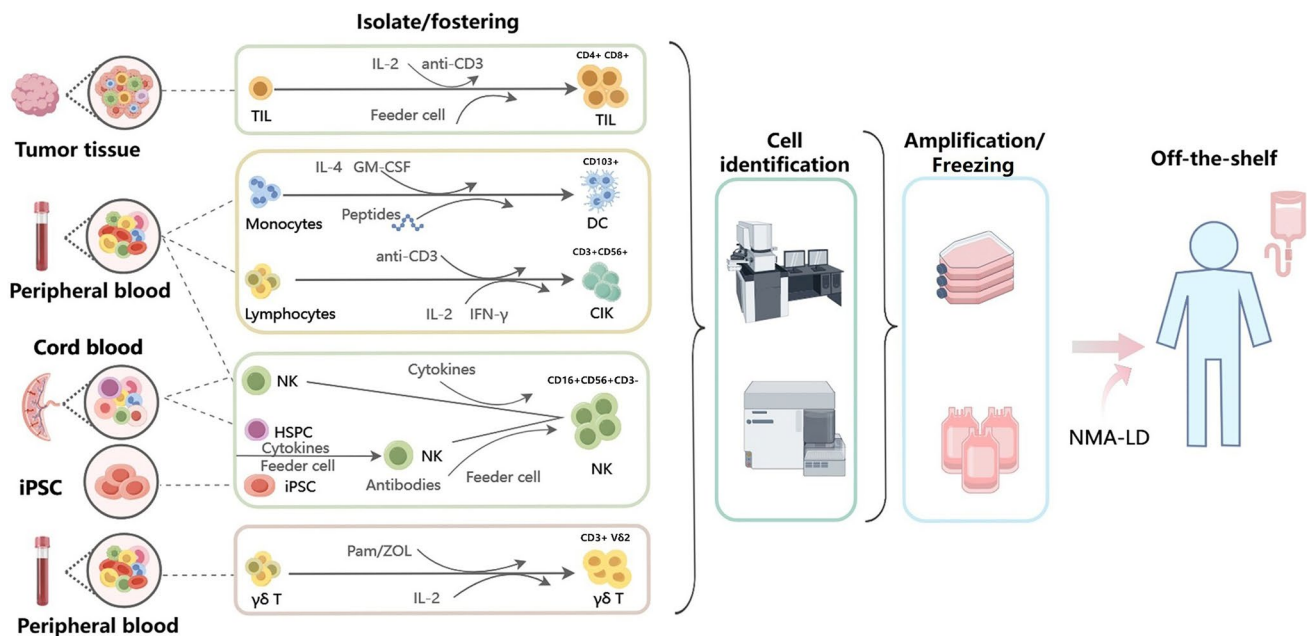


Fig. 3 Process of non-genetically modified adoptive cell therapies. TIL: Tumor tissue is obtained from the subcutis and lymph nodes, digested or fragmented tumor tissue is cultured in an IL-2-rich medium, OKT3, allogeneic feeder cells, and IL-2 are added to stimulate T-cell proliferation, and concentrated TIL cells are harvested; CIK: Lymphocytes from the patient's blood were extracted, and CIK cells were cultured in vitro and expanded in large quantities by sequentially adding IFN- γ , OKT3, and rhIL-2; NK: use of lymphocytes, NK-92, iPSC and other cells derived from the patient's peripheral blood or umbilical cord blood, NK cells obtained by in vitro

cultivation and mass expansion in media enriched with a variety of cytokines and feeder cells; $\gamma\delta$ T: lymphocytes from the patient's peripheral blood were extracted, and $\gamma\delta$ T cells were obtained by sequential addition of ZOL, Pam, and IL-2 for in vitro culture and bulk expansion. Finally, the cell product is reinfused back into the patient after they have received non-cleared lymphocyte clearance. TILs, tumor-infiltrating lymphocytes; CIK, cytokine-induced killer cells; OKT3, anti-CD3 monoclonal antibody; IL-2, interleukin-2; DC, dendritic cells; NK: natural killer cells; IFN- γ : interferon- γ ; iPSC: induced pluripotent stem cells; By Figdraw

number of TILs infused. A greater number correlates with a significant clinical response. To improve the efficacy and durability of TIL therapy, it is essential to select TIL subsets that are appropriate for expansion and exhibit high response rates. The proportion of CD8⁺ T cells in infused TILs is significantly associated with clinical response; however, CD4⁺ T cells may also induce antitumor activity, recruit and enhance CD8⁺ T-cell antitumor activity, or directly mediate antitumor effects by secreting Th1 cytokines [51, 52].

The optimal CD8⁺ T-cell ratio requires further investigation. Another approach involves combining TIL therapy with ICIs, as the upregulation of CTLA-4 and PD-1 molecules on effector T cells interacts with B7-1/B7-2 and PD-L1 on antigen-presenting or tumor cells, resulting in T-cell dysfunction. This inhibition can be mitigated using anti-CTLA-4 and anti-PD-1 antibodies, hence enhancing T-cell toxicity. The IOV-COM-202 trial (NCT03645928) is an open-label, prospective, multicenter, multi-cohort phase II study that demonstrated an ORR of 47.1% among patients with metastatic NSCLC who had not previously received ICI therapy when treated with lifileucel combined with pembrolizumab. In patients with EGFR wild-type and PD-L1-negative tumors, ORR was significantly higher, with four of eight

patients with EGFR wild-type negative PD-L1 achieving a response [53]. Furthermore, a multicenter phase III trial is ongoing to evaluate LN-145 TIL therapy, with or without pembrolizumab, in patients with unresectable or metastatic melanoma, advanced, recurrent, or metastatic head and neck squamous cell carcinoma, and locally advanced or metastatic NSCLC [54].

TIL therapy requires the acquisition of sufficient tumor samples from patients, a challenge that is often encountered with various tumor types. Patient selection is critical for the efficacy of TIL therapy. Radvanyi et al. found that tumors with $\geq 10\%$ CD8⁺ TIL infiltration exhibited significantly higher response rates (ORR = 45% versus 12%, $P < 0.01$), whereas TIL therapy demonstrates enhanced efficacy against minimal residual disease, including post-resection melanoma [51]. The culture procedure of TILs from tumors lacks strict scientific standards, hence complicating quality control. Gohen et al. reported that new antigen-specific T cells may be extracted from the peripheral blood lymphocytes (PBLs) of patients [55, 56]. Li et al. developed a method to harvest and rapidly proliferate new antigen-specific T cells from PBLs of patients with PD-1⁺. This method was employed to treat four patients who had progressed after

multiple lines of treatment and developed acquired resistance to anti-PD-1 therapy. T cells were proliferated with this method and administered with anti-PD-1 antibodies for treatment, yielding objective responses in three patients [57]. Based on these results, related clinical trials are underway (ClinicalTrials.gov identifier: NCT04268108), and this approach may serve as an alternative to TIL therapy for the treatment of more patients with cancer. Moreover, the TIL production cycle is long, presenting a challenge for patients with rapidly progressing tumors. Lovance has developed an innovative, simplified expansion protocol (Gen-2) that reduces the expansion duration to 22 days. This new protocol produces TILs with enhanced antitumor efficacy *in vitro* compared to endogenous TILs [58]. Additionally, the inclusion of diverse cytokines in the culture protocol may have varying effects on the efficacy of TIL therapy. Although TIL therapy has achieved FDA approval for advanced melanoma, its efficacy in epithelial cancers remains inconsistent. Recent single-cell investigations indicate that stem-like TCF7⁺ CD8⁺ TILs exhibit superior persistence post-infusion, whereas terminally exhausted subsets correlate with relapse. This underscores the need for novel TIL protocols to preserve the precursor phenotype and to isolate and increase stem-like neoantigen-specific T cells, potentially facilitating the development of more efficacious T-cell-based immunotherapies [59]. We contend that TIL expansion should integrate metabolic priming (mannose supplementation to enhance oxidative phosphorylation) and spatial targeting (coupling with localized radiotherapy to enhance homing). A fundamental challenge in TIL therapy is the insufficient understanding of the specific tumor antigens recognized by the infused T-cell repertoire in each patient. TILs are presumed to be enriched for tumor reactivity, yet they represent a polyclonal mixture that targets both neoantigens from somatic mutations and common tumor-associated antigens. Advances in high-throughput TCR sequencing and neoantigen identification are beginning to elucidate this complexity. Correlating the TCR clonality of the infused product with clinical response may help identify the truly tumor-reactive clones and inform the selection of more effective TIL products [55–57].

CIK and DC–CIK

CIK cell therapy is an immunotherapeutic strategy that utilizes lymphocytes derived from the patient's blood. The cells are expanded and activated *in vitro* by sequentially adding IFN- γ , anti-CD3 antibody (OKT3), and high-dose recombinant human IL-2 (rhIL-2) before being reinfused back into the patient to enhance the antitumor effect (Fig. 3). CIK cells are a heterogeneous population of cells exhibiting T lymphocyte and NK cell characteristics, marked by CD3⁺ and

CD56⁺ expression, and have significant antitumor activity and extensive therapeutic potential [60, 61].

CIK cells recognize and kill tumor cells through several mechanisms (Fig. 2). CIK cells possess polyclonal T-cell receptors (TCRs), enabling them to recognize tumors in an MHC-restricted manner. Furthermore, CIK cells exhibit NK receptors (NKG2D, NKp30, and DNAM-1), facilitating their ability to recognize tumors in an MHC-unrestricted manner. NKG2D is a crucial molecule in facilitating the antitumor activity of CIK cells [61]. CIK cells can directly kill tumor cells through granzyme- and perforin-mediated cytotoxicity and secrete cytokines, such as IFN- γ , TNF- α , and IL-2, to modulate the immune response, indirectly killing tumor cells [62]. Moreover, CIK cells can induce apoptosis in tumor cells by binding to the Fas receptor on the tumor cell membrane, activating programmed cell death.

CIK cell therapy has been extensively employed to treat various solid tumors, including liver, lung, breast, colorectal, nasopharyngeal, gastric, and renal cancers [63–69]. CIK cell treatment demonstrates notable efficacy in patients post-surgery or chemoradiotherapy [70]. It can eliminate residual minimal metastatic lesions, prevent the dissemination and recurrence of cancer cells, and enhance immunity [70]. Liu et al. conducted a multicenter, open-label, randomized-controlled phase II clinical trial to evaluate the effect of CIK cell therapy combined with chemotherapy on advanced squamous NSCLC; the addition of CIK therapy to chemotherapy extended PFS by 4.7 months and OS by 10.7 months, and the addition of CIK cell therapy did not result in an increase in adverse events (AE) [71]. In a study involving 85 patients with primary hepatocellular carcinoma (HCC), CIK cell therapy was administered alongside transarterial chemoembolization (TACE) and radiofrequency ablation. The findings revealed that the recurrence-free survival rate of the CIK treatment group was 68.9%, significantly higher than that of the control group (20.0%) [72]. In a clinical trial of CIK cell therapy combined with first-line chemotherapy for patients with unresectable colorectal cancer, the combined treatment significantly enhanced OS and PFS, with the median overall survival increasing from 24.1 to 54.7 months and median disease-free survival (DFS) increasing from 14.6 to 25.7 months [73]. Numerous randomized-controlled trials (RCTs) have substantiated the clinical efficacy of CIK cell therapy, especially in the adjuvant treatment of HCC. According to the National Comprehensive Cancer Network (NCCN) guidelines, the pivotal phase III clinical trial conducted by Lee et al. demonstrated that patients with HCC receiving postoperative adjuvant autologous CIK cell therapy ($\geq 5 \times 10^9$ cells per infusion, six courses) exhibited significantly prolonged median recurrence-free survival (RFS) compared to surgery-alone controls (44.0 versus 30.0 months, $P=0.01$), with a 21% enhancement in 5-year overall survival (OS) (66% versus 45%, $P=0.008$).

Subgroup analysis indicated that patients receiving ≥ 9 CIK infusions achieved significantly greater survival benefits (hazard ratio = 0.59, 95% confidence interval [CI] 0.43–0.81) without an increased risk of grade ≥ 3 AEs [68]. This study offers high-level evidence supporting standardized CIK protocols in HCC management and constitutes the exclusive CIK-related recommendation included in NCCN guidelines. Moreover, clinical studies have indicated that patients with positive PD-L1 expression may demonstrate an improved response to CIK-based combination therapies [67].

CIK cells are typically derived from autologous peripheral blood mononuclear cells (PBMCs). However, patients with cancer with poor immune reactivity often fail to yield sufficient antitumor lymphocytes, and older adult patients may struggle to endure repeated blood draws. Umbilical cord blood and allogeneic sources offer ‘off-the-shelf’ potential. Preclinical studies revealed that allogeneic CIK cells exhibit cytotoxic activity without inducing graft-versus-host disease (GvHD) and demonstrate enhanced antitumor efficacy [74]. Clinical studies have demonstrated that among 15 patients treated with CB-CIK cell therapy, one patient with HCC and one patient with esophageal cancer achieved a complete response (CR), two patients with ovarian cancer attained partial response (PR), ten maintained stable disease (SD), and one patient with HCC experienced progressive disease (PD). The observed safety and feasibility profile indicates that CB-CIK may be a promising therapeutic option [75]. Another study demonstrated that combining CB-CIK with second-line chemotherapy markedly enhanced PFS and median survival in patients with advanced solid tumors who had progressed following first-line chemotherapy [76]. Current research on allogeneic CIK cells is limited, and their scalability is hindered by protocol variability. Further research is required to investigate the efficacy and safety of allogeneic CIK cells.

DC–CIK therapy is an immunotherapy approach that combines DCs and CIKs. This therapy employs the antigen-presenting ability of DCs and the cytotoxicity of CIKs to collaboratively target tumor cells, therefore enhancing the body’s antitumor immune response. DC–CIK cells recognize antigens on the surface of tumor cells and stimulate T-cell responses, whereas CIK cells directly kill tumor cells. The coculture of DC and CIK cells enhances the antitumor efficacy of CIK cells, rendering them more lethal. The interaction between DC and CIK cells modifies the expression of surface molecules in both cell types, resulting in increased secretion of IL-12 and enhanced cytotoxic activity [77]. DC–CIK therapy has been employed in various solid tumor treatments, including NSCLC, colorectal, liver, and pancreatic cancers [78–82]. Studies have demonstrated that DC–CIK therapy can enhance the effect of chemotherapy, bolster the patient’s immune response, and diminish the risk of tumor recurrence. A meta-analysis encompassing

12 RCTs involving 826 patients with solid cancer demonstrated that DC–CIK treatment combined with chemotherapy significantly improved OS, DFS, and ORR compared to chemotherapy alone [83]. Another meta-analysis comparing 10 CIK monotherapy trials with 16 DC–CIK combination therapy trials revealed that the DC–CIK group did not achieve statistically higher objective response rates (ORR) relative to the CIK monotherapy group (42.5% versus 38.1%, $P=0.12$). This seemingly counterintuitive outcome—where the theoretically improved DC–CIK failed to exhibit significant therapeutic benefit—can be attributed to three principal factors: non-standardized antigen loading protocols in DC–CIK preparation: In specific trials, DCs were insufficiently loaded with tumor-specific antigens, undermining their antigen-presenting efficacy. Patient population heterogeneity: CIK monotherapy trials primarily enrolled postoperative patients with minimal residual disease (demonstrating significant extension of median recurrence-free survival to 44.0 versus 30.0 months in HCC adjuvant therapy), whereas DC–CIK trials included patients at more advanced stages. The severely immunosuppressive TME in advanced disease likely inhibited immune-cell activation. Variability in cellular expansion efficiency during DC–CIK production: The percentage of CD3⁺CD56⁺ cells—essential effector elements in DC–CIK products—demonstrated significant inter-study fluctuations (range: 15–40%). This manufacturing variability may directly affect therapeutic efficacy, reproducibility, and cross-trial comparability [66]. DC vaccines are an emerging immunotherapy, and tumor interventional antigens loaded onto DCs include peptides, recombinant proteins, and entire tumor cells or their lysates. The diversity of DC–CIK regimens, encompassing the specific tumor antigen utilized, requires additional clinical trials to validate the efficacy of DC–CIK and CIK cell therapy in the treatment of solid tumors [60]. DC enhances the immune response of CIK cells by enhancing IL-12 secretion [84]. The benefit of DC–CIK therapy lies in its non-MHC restriction, allowing for broad applicability across diverse patient groups. Moreover, ex vivo multiplication and activation of CIK cells is comparatively easy and cost-effective, making it a cost-effective treatment option. DC–CIK therapy may be utilized in combination with other treatment methods, including chemotherapy, radiotherapy, and targeted therapy, to improve the treatment effects [85, 86]. In a clinical phase III trial, a new PD-1 blocking-activated DC–CIK cell PD1-T was developed, and PD1-T cells were added to first-line XELOX combined with bevacizumab [87]. The study demonstrated that in previously untreated patients with metastatic colorectal cancer, the combination of PD1-T cell treatment plans significantly improved PFS and OS, with good tolerance [87].

Although CIK cell therapy has demonstrated therapeutic potential in several clinical studies, it has some limitations.

A study demonstrated that alterations in the ratio of CD4⁺ T cells, regulatory T cells (Tregs), and CD3⁺CD56⁺ T cells in an in vitro-expanded CIK cell population significantly affect the antitumor activity of CIK cells [88]. The Th1/Th17 cell subpopulation of CD4⁺ T cells significantly contributes to CIK therapy by inhibiting the expression of CIK inhibitory receptors [66]. It enhances the antitumor efficacy of CIK cells through IL-17A and promotes the migration and motility of CIK cells through the IFN- γ /CXCL9-11/CXCR3 pathway; however, the impact of CIK cell expansion may vary among patients [88]. Currently, most clinical trials have insufficient quantification of CIK cells, and owing to variations in culture protocols, there is significant heterogeneity in CIK cells among different studies. Additional study is required to identify the main effector cell subsets responsible for CIK cell infiltration into tumors and for mediating cytotoxicity. The antigenic specificity of CIK cells remains less defined compared to TILs. Their polyclonal TCR repertoire and activation through NKG2D provide extensive, MHC-unrestricted tumor recognition. However, this complicates the identification of the specific tumor ligands responsible for their cytotoxicity. Future research utilizing ligand screening platforms and single-cell analyses may elucidate the predominant antigens recognized by CIK cells, thereby facilitating the development of more targeted and effective CIK products.

NK Cells

NK cells are innate lymphocytes that recognize and kill virus-infected and tumor cells through direct cell-mediated cytotoxicity and the secretion of cytokines and chemokines, thereby sharing effector functions with conventional $\alpha\beta$ T cells, particularly CD8 cytotoxic T lymphocytes (CTL) [89]. NK cells derive from CD34⁺ common lymphoid progenitor cells, with a half-life of approximately 7–10 days, and are integral to immune regulation through interactions with DC [86, 90]. The activity of NK cells is modulated by several activating and inhibitory receptors expressed on the cell surface, enabling the rapid secretion of various cytokines and chemokines, the recruitment of additional immune cells, and the enhancement of the adaptive immune response of T cells and B cells.

NK cells comprise 10–15% of human peripheral blood lymphocytes and are generally identified by the CD16⁺CD56⁺CD3⁺ phenotype [91]. Their primary features are the synthesis of type 1 cytokines, including IFN- γ and TNF. Most healthy cells express type I major MHC-I, which stimulates robust inhibitory signals to inhibit NK cell activities. Tumor cells often escape recognition by CD8-positive cytotoxic T cells by diminishing MHC-I expression, and the loss of MHC-I ligands and/or the expression of ligands for activating receptors will change the balance of signaling

pathways within NK cells, resulting in the activation of NK cells and attack of target cells [92]. NK cells participate in cancer immune surveillance, and compared with T cells, NK cells are part of the innate immune system and can respond more rapidly and kill abnormal cells [93]. Second, NK cells do not depend on MHC molecules to recognize target cells and can recognize and kill a wider range of tumor cells. NK cells can kill tumor cells through several mechanisms (Fig. 2). First, NK cells can directly kill tumor cells by secreting perforin and granzyme. Additionally, NK cells can kill tumor cells through antibody-dependent cytotoxicity (ADCC). Cytokines produced by NK cells can modulate immune responses and enhance the activity of other immune cells [94]. In NK cell therapy, autologous NK cells may inadequately respond effectively to hematologic or solid cancers owing to the inhibitory effect of interactions with their MHC I molecules, whereas allogeneic NK cells achieve persistent infiltration through killer immunoglobulin-like receptors (KIR) ligand mismatch [95]. Since many patients with advanced cancer cannot remove tumors to obtain TILs, the administration of allogeneic T-cell infusions carries the risk of GvHD, whereas allogeneic NK cells are safe [96]. Miller et al. treated patients with Hi-Cy/Flu induction therapy and infused expanded allogeneic NK cells. Adoptively transferred NK cells demonstrated in vivo amplification and did not induce GvHD, exhibiting a low incidence of CRS [21].

Studies indicate that the molecular level and subtypes of NK cells correlate with the prognosis of patients with gastrointestinal stromal tumors, colorectal cancer, and breast cancer [97–100]. Based on the characteristics of NK cells, utilizing NK cells for adoptive cell therapy to treat solid tumors has great potential.

Hong et al. treated 48 patients with advanced lung adenocarcinoma with or without epidermal growth factor receptor (EGFR) mutations with afatinib combined with or without autologous NK cell therapy. The results indicated that in patients with EGFR mutation-positive tumors, the targeted drug afatinib combined with autologous NK cell therapy significantly increased the response rate [101]. ParkHurst et al. treated eight patients with metastatic melanoma or renal cell carcinoma who were unsuitable for TIL therapy using non-myeloablative chemotherapy followed by adoptive transfer of autologous NK cells. Although no objective clinical response was observed, the transferred NK cells remained for 1 week to several months [102]. In a phase I clinical trial, Sakamoto et al. administered induced T-cell-stimulated, peripheral blood mononuclear cell (PBMC)-expanded autologous NK cells to patients with unresectable, locally advanced, or metastatic gastrointestinal tumors. Autologous NK-cell therapy demonstrated a favorable safety profile, with an approximately twofold increase in peripheral blood cytotoxicity [103]. Despite these safety advantages, autologous NK cell therapy faces significant challenges in

solid tumor treatment. The TME induces NK cell exhaustion, impairing their cytotoxicity. Furthermore, autologous MHC-I molecules deliver inhibitory signals through KIR receptors, impairing NK-cell-mediated tumor recognition. NK cell frequency and proliferative capacity in peripheral blood are also reduced in patients with advanced cancer [92, 104, 105]. Allogeneic NK cell therapy offers a strategy to overcome these barriers through KIR ligand mismatch. The therapeutic principles were first established in hematologic malignancies. For instance, in acute myeloid leukemia, KIR-mismatched allogeneic NK infusions achieved a 47% complete remission (CR) rate, compared with the 5% CR rate observed in autologous NK therapy [21]. These findings provide a strong rationale for applying allogeneic NK strategies to counteract the inhibitory autologous MHC-I environment in solid tumors. Currently, allogeneic NK cells are primarily used to avoid changes in NK cells. In addition to peripheral blood, NK cells can be derived from umbilical cord blood, NK-92, and induced pluripotent stem cells [106–109]. In a study by Liang et al., 36 patients with recurrent breast cancer were treated with either autologous or allogeneic NK-cell therapy. The adverse reactions were similar between groups; however, patients receiving allogeneic NK therapy exhibited increased total numbers of T cell and NK cell counts, experienced greater reductions in tumor diameter and computed tomography (CT) attenuation value compared with those treated with autologous NK cells [110].

Adoptive NK cell therapy must overcome three major barriers: limited durability of clinical responses, suppression mediated by cytokines/chemokines within the TME, and insufficient long-term immunologic memory. Recent studies demonstrate that IL-15 signaling sustains NK cell function through an XBP1s-PIM-2 positive-feedback loop that regulates survival and effector activity [111, 112]. To address these limitations, engineered NK cell conjugates have demonstrated enhanced antitumor efficacy by simultaneously engaging tumor-specific surface antigens and co-activating NK receptors (CD16 and NKG2D), thereby stimulating potent cytotoxic responses against solid tumors [105]. Genetic editing and molecular engineering approaches may further mitigate these barriers, and modification of NK cells to express CAR can substantially enhance their tumor-targeting ability.

Furthermore, iNKT cells are a distinct subset of T lymphocytes characterized by expression of semi-invariant T-cell receptor (TCR), composed of V α 24-J α 18 chains in humans. These cells recognize specific lipid antigens presented by CD1d molecules and express activating and inhibitory NK receptors, thereby bridging innate and adaptive immunity [113]. Additionally, iNKT cells develop in the thymus and contribute to immunosurveillance of tumor cells in the peripheral blood. Activation occurs through recognition of CD1d-presented antigen in an MHC-independent

manner, enabling rapid cytokine secretion and subsequent activation of multiple immune effector populations [114]. Following optimal activation, iNKT cells can differentiate into highly functional effector subsets, with some cells surviving and acting as effector memory cells. Johan et al. conducted a prospective study and found that low concentrations of circulating iNKT were related to poor prognosis in patients with head and neck squamous cell carcinoma [115]. Rebuilding the gradient of circulating iNKT concentrations may enhance patient survival rates. In a phase I clinical trial involving patients with advanced or recurrent non-small cell lung cancer, intravenous infusion of autologous iNKT cells was well tolerated, with no severe adverse reactions. Infused iNKT cells accumulated in lung cancer lesions, demonstrating evidence of an antitumor response [22]. Similarly, Exley et al. infused autologous ex vivo expanded iNKT cells into patients with stage IIIB-IV melanoma, with a good safety profile, increased activity and number of iNKT in circulation, and potential antitumor effects.

$\gamma\delta$ T Cells

Adoptive transfer of $\gamma\delta$ T-cell therapy represents an innovative cancer treatment strategy that harnesses the patient's immune system to recognize and eliminate cancer cells. $\gamma\delta$ T cells are a unique T-cell subset that recognizes non-peptide antigens through their distinct T-cell receptors (TCR) and play a key role in tumor immune surveillance [116]. These cells mediate multiple antitumor mechanisms, including direct cytotoxicity, cytokine-mediated killing, facilitation of innate immune response, and production of IL-17. Direct cytotoxicity involves ADCC, perforin–granzyme-mediated killing of tumor cells, and induction of apoptosis through FasL and TRAILR receptors. $\gamma\delta$ T cells also target several solid tumors through natural killer cell receptors (NKR) and by specifically recognizing phosphate metabolites such as isopentenyl pyrophosphate. Cytokine-mediated tumor control is achieved through the secretion of cytokines, including IFN- γ , TNF- α , and IL-2, which promote the proliferation and differentiation of CD8⁺ T cells and enhance cellular antitumor activity [117, 118]. Notably, infiltrating $\gamma\delta$ T cells have been identified as the strongest positive prognostic indicator in 25 cancers [119].

The application of $\gamma\delta$ T-cell therapy in oncology is primarily reflected in two approaches: adoptive immunity, in which activated $\gamma\delta$ T cells are transferred to severe combined immunodeficiency (SCID) mice to inhibit tumor growth, and directly activating $\gamma\delta$ T cells in SCID mice with activators, including N-PBs and IL-2 [120]. These foundational studies establish a theoretical rationale for the clinical translation. In clinical studies, $\gamma\delta$ T-cell therapy has been employed to treat various cancer types, including solid tumors and hematological tumors [121]. For instance, $\gamma\delta$ T cells are expanded

with phosphorylated antigens or anti- $\gamma\delta$ TCR antibodies and transferred to patients to enhance the body's antitumor response (Fig. 3) [122–124]. Additionally, $\gamma\delta$ T cells can enhance DC maturation by secreting cytokines, including IFN- γ and TNF- α , thereby activating adaptive immune responses. In a study of patients with metastatic solid tumors, ex vivo expansion-return autologous V γ 9V δ 2 T-cell therapy combined with conventional therapies was feasible, well-tolerated, and exhibited antitumor activity [125]. Benouna et al. similarly demonstrated that autologous $\gamma\delta$ T-cell immunotherapy was safe in patients with advanced renal cell carcinoma, well-tolerated, and exhibited antitumor effects [126]. Some patients with cancer have impaired $\gamma\delta$ T-cell function and difficulty in expanding circulating V δ 2 T cells for autologous immune-cell therapy. The unique MHC-independent recognition of $\gamma\delta$ T provides an advantage in reducing allogeneic rejection, making allogeneic adoptive $\gamma\delta$ T cell transfer a promising therapeutic strategy. Allogeneic V γ 9V δ 2 T-cell immunotherapy has good clinical safety and can extend the survival of patients with advanced lung or liver cancer [127].

Despite the potential of $\gamma\delta$ T-cell therapy in antitumor treatment, there are still some challenges. First, the expansion of $\gamma\delta$ T cells in vitro is limited, and optimized manufacturing protocols are lacking. Currently, the most effective method involves phosphoantigen stimulation or anti- $\gamma\delta$ TCR antibodies [128]. Using bispecific molecules to selectively recruit resident $\gamma\delta$ T cells and connecting $\gamma\delta$ T cells to tumor cells expressing epidermal growth factor receptor (EGFR) is a new direction of research [129]. Second, $\gamma\delta$ T cell's heterogeneity causes their variable therapeutic efficacy among patients, complicating treatment standardization. In addition, more severe immune-related AEs, including cytokine release syndrome, have also been reported, and the long-term effects and safety of $\gamma\delta$ T-cell therapy require further clinical investigation. Notably, the clinical efficacy of $\gamma\delta$ T-cell therapy remains inconclusive despite preclinical promise. A phase II trial in metastatic solid tumors reported an ORR of 18%, highlighting the need for large-scale phase III trials to validate its role in adjuvant settings [125].

Representative Active Trials of Non-Genetically Modified Adoptive Cell Therapies

Non-genetically modified adoptive cell therapy has emerged as a promising frontier in immunotherapy that enhances antitumor or immunoregulatory functions of immune cells without altering their genetic material. Current clinical trials are exploring innovative approaches, including cytokine preconditioning and combination therapies, to optimize the therapeutic effects of T cells, CIK cells, NK cells, and $\gamma\delta$ T cells. These trials aimed to improve safety, broaden accessibility, and avoid risks associated with gene editing,

including off-target effects and insertional mutations (Supplementary Table 1). Current non-gene-edited adoptive cell therapy (ACT) clinical trials are undergoing three major paradigm shifts. The first is biology-driven design, optimizing expansion protocols based on inherent cellular properties: TIL therapy emphasizes the expansion of tumor-reactive T cells, often combined with IL-2 to maintain activity (7 trials). DC-CIK leverages DCs to enhance antigen presentation (NCT06015269); memory-like NK (CIML) enhances persistence through cytokine pre-activation (NCT06318871). $\gamma\delta$ T cells are engineered, with fusion constructs such as PD-1 antibody-linked $\gamma\delta$ T cell ($\gamma\delta$ T-PD-1 Ab), to enhance tumor targeting. The second shift involves innovative combination strategies, used in 82% of trials (28/34), including ICIs (PD-1 antibody with NK cells, NCT05429697), targeted drugs (HER2 inhibitors with NK therapy for breast cancer, NCT05385705), and local therapies (cryoablation synergizing with DC-CIK for liver cancer, NCT05622825) to overcome TME suppression. Breakthroughs in allogeneic and standardized approaches are evident, marked by a significant increase in allogeneic therapies (NK: 81%, $\gamma\delta$ T: 100%), the development of "off-the-shelf" products with shortened manufacturing time (2–3 weeks), and the implementation of cell quality thresholds in 27 trials (79%), such as requiring CD3⁺CD56⁺ $\geq$ 30%. Future prospects focus on three key areas: solid TME remodeling, intelligent manufacturing, and advanced engineering. With therapies such as lifileucel gaining approval, ACT is transitioning from salvage therapy for late-stage disease toward earlier lines of treatment, including neoadjuvant application, ultimately aiming for curative therapy in solid tumors.

Conclusions and perspectives

Adoptive immune-cell therapies have advanced significantly, with multiple FDA-approved products for hematological cancer treatment [130]. In solid tumors, TIL therapy, particularly lifileucel, has received FDA approval for the treatment of patients with advanced melanoma who progressed after ICIs and targeted therapies [29]. Combined strategies involving non-genetically modified adoptive cell therapy with ICIs have exhibited tremendous potential in enhancing treatment efficacy. For instance, TIL therapy combined with the anti-PD-1 antibody, pembrolizumab, has exhibited clinical efficacy in patients with melanoma and non-small cell lung cancer by alleviating TME-mediated immune suppression and enhancing the activity and persistence of immune cells, thereby enhancing therapeutic outcomes [48, 131]. The combination of targeted therapies against specific molecular targets with adoptive cell therapy allows for treatment tailored to the specific biological characteristics of the tumor. For instance, combining anti-HER2 antibody–drug

conjugates (ADC) with NK cell therapy has demonstrated synergistic effects among patients with HER2-positive breast cancer [132]. The combination of targeted therapies with adoptive cell therapy targets specific tumor cell markers and enhances the body's immune response through adoptive cell therapy. Furthermore, local treatments, including radiotherapy and TACE combined with adoptive cell therapy, can enhance local tumor control and stimulate systemic immune responses (133). Developing genetic engineering technologies has provided novel approaches for over-the-counter cellular treatments. Despite the remarkable efficacy of CAR-T cell therapy in hematological tumors, their implementation in solid tumors still faces challenges, including targeting specific targets, non-tumor tissue toxicity, and difficulty in treating solid tumors [134–136] (Table 1). The selection of these cell therapies is context-dependent, and future directions may see a convergence, such as arming non-genetically modified cells with cytokines or combining them with genetic modifications to overcome specific barriers.

We compared four types of non-genetically edited adoptive cell therapies (Table 2). One of the challenges of non-genetically modified adoptive cell therapy is the difficulty in increasing immune cells in vitro. Prolonged treatment regimens can hinder patients from advancing quickly and receiving effective cell therapy. Xiao et al. and Boucher et al. demonstrated that ex vivo stimulation of donor-derived $\gamma\delta$ T cells utilizing zoledronic acid, IL-2, and genetically engineered K562-based APCs results in substantial massive expansion (633-fold in Boucher et al.), with a high proportion exhibiting an effector memory phenotype (43%), indicating potential for robust, long-lasting antitumor responses [137, 138]. Non-gene-edited ACT therapies exhibit distinct advantages in solid tumor

treatment, including a higher safety profile and multi-target killing capabilities. However, its clinical applications encounter several challenges: inadequate standardization of in vitro expansion protocols, individual heterogeneity of efficacy, and absence of biomarkers to predict ACT persistence or TME-mediated resistance.

To overcome these obstacles, future research needs to focus on three major directions:

- Combination therapies: Exploring synergy with ICIs and metabolic modulators. ACT is combined with ICIs and spatial multi-omics, dynamically analyzing the mechanisms of TME remodeling through single-cell RNA sequencing (scRNA-seq) and spatial proteomics (CODEX) to precisely identify intervention targets for combination therapies.
- Standardization of expansion protocols: Enhancing consistency in vitro culture techniques. Developing a unified efficacy evaluation system, replacing conventional metrics such as cell counts or phenotype ratios (CD3/CD56) with metabolic functional scoring (oxidative phosphorylation/ATP flux) and utilizing patient-derived organoid killing models to objectively measure the in vivo activity of ACT products—and exploring the potential of ACT as a neoadjuvant therapy, with complete pathological response and circulating tumor DNA (ctDNA) clearance as endpoints, to confirm the potential of activating systemic immunity through preoperative ACT. By thoroughly combining mechanistic research with advanced technologies (spatial histology and organoid platforms), ACT holds potential for transitioning from empirical cell infusion into a precision immunotherapy, accelerating its bench-to-bedside translation.

Table 1 Comparison of non-genetic editing ACT and modified cell therapy

Clinical therapy	Non-genetic editing ACT	Modified cell therapy
Representative types	TILs, CIK cells, NK cells, $\gamma\delta$ T cells	CAR-T, TCR-T
Targets	Targeting multiple tumor antigens	Targeting specific tumor antigens, CAR-T: MHC-independent; TCR-T: MHC-restricted
Raw material	Excised tumor tissue/peripheral blood mononuclear cells obtained by leukapheresis	Peripheral blood mononuclear cells obtained by leukapheresis
Manufacturing time	3–4 weeks	2–3 weeks
Toxicity	Generally well-tolerated, with rare effects on non-neoplastic tissues	Non-tumor tissue targeting effects with severe adverse events (cytokine release syndrome)
FDA Approval	Lovance's lifileucel (AMTA-GVI) has received FDA approval for advanced melanoma	Several products have received FDA approval, including ABECMA, BREYANZI, CARVYKTI, KYMRIAH, TECARTUS, and YESCARTA
Clinical application	The most effective against melanoma, also effective in non-small cell lung cancer, breast cancer, and colorectal cancer	CAR-T mainly treats blood cancers; TCR-T mainly treats solid tumors
Advantage	Utilizing the patient's own immune cells for personalized therapy, leveraging polyclonal advantages	Specific epitope recognition, high affinity, MHC-independent
Disadvantage	Complex manufacturing, high costs, tumor tissue dependence, and insufficient expansion	Poor infiltration, short durability, and prone to antigen escape ^(137–139)

Table 2 Key features of major non-genetically modified adoptive cell therapies for solid tumors

Therapy type	TILs	CIK cells	NK cells	$\gamma\delta$ T cells
Primary source	Resected tumor tissue	Peripheral blood lymphocytes	Peripheral blood, cord blood, iPSC, NK-92 cell line	Peripheral blood lymphocytes
Key marker phenotype	CD3 ⁺ (with diverse subtypes)	CD3 ⁺ CD56 ⁺	CD3 ⁺ CD56 ⁺ CD16 ⁺	TCR $\gamma\delta$ ⁺ (V γ V δ 2 subset common)
Recognition mechanism	MHC-restricted TCR recognition of tumor-associated antigens and neoantigens	TCR ⁺ NKG2D	NKG2D ⁺ KIR	TCR ⁺ NKR ^s
Major advantages	High tumor specificity and presence of neoantigen-reactive clones	MHC-unrestricted killing and broad antitumor activity	No risk of GVHD in allogeneic setting and "off-the-shelf" potential	MHC-independent tumor recognition and bridges innate and adaptive immunity
Major challenges	Difficult to obtain sufficient tumor tissue and limited efficacy in some epithelial cancers	Lack of standardized protocols, variable CD3 ⁺ CD56 ⁺ ratios, and unclear dominant effector subset	Suppressive TME induces exhaustion and limited persistence in vivo	Limited in vitro expansion and inconclusive clinical efficacy in trials

- Biomarker-based stratification: Establishing predictive biomarkers to guide ACT use. Current biomarkers (PD-L1 expression, TIL density) fail to represent the dynamic efficacy of ACT in real-time. However, in TIL therapy, high TCR diversity in the infused product is associated with poorer outcomes, as it contains a significant fraction of non-tumor-reactive clones. Therefore, validation through in vitro coculture assays focusing on the tumor specificity of high-frequency clones and dynamic monitoring of the circulating TCRomic repertoire can be performed.

The combination of non-genetically modified adoptive cell therapy with ICIs, targeted therapies, and local treatments offers novel strategies for solid tumor treatments. The advancement of genetic engineering offers novel possibilities for adoptive cell therapies. Future research should focus on improving the efficacy of adoptive cells, addressing immune suppression in the TME, and optimizing combination therapy strategies.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00262-025-04274-y>.

Acknowledgements Not applicable.

Author Contributions Qinghuizi Luo: Conceptualization, Visualization, Formal analysis, Writing—original draft. Wendi Jiang: Conceptualization, Visualization, Formal analysis. Yiyang Xie: Conceptualization, Visualization, Formal analysis. Zhe Dai: Investigation. Jialong Zhu: Investigation. Yuanyang Du: Investigation. Xiaoyue Tao: Investigation. Zengjie Lei: Supervision, Funding acquisition. Xiaoyuan Chu: Supervision, Funding acquisition. Bingji Wang: Conceptualization, Writing—review&editing. Gongbo Fu: Conceptualization, Supervision, Funding acquisition, Writing—review&editing. All authors reviewed the manuscript.

Funding This work was supported by the National Natural Science Foundation of China 82002591 (to Gongbo Fu), 82072725 (to Xiaoyuan Chu), the Key Program of Jiangsu Provincial Health Commission ZD2021039 (to Xiaoyuan Chu), and the China Postdoctoral Science Foundation 2020M670090ZX (to Zengjie Lei) and 2021MD703958 (to Gongbo Fu).

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethics Approval Not applicable.

Consent for Publication Not applicable.

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